Project Field Visit Report: Investigation & Optimization of the Tea Withering Process

Analysis of Current Trough Operations and Proposed Automated Air Mixing System

Location:	St. Coombs Estate Factory, Talawakelle
Date of Visit:	May 2026
Focus Area:	Withering Trough Dynamics and Automation Potential
Prepared For:	Project Review Committee / Tea Research Institute

1. Introduction & Overview of Factory Operations

St. Coombs Estate, situated in Talawakelle, serves as the prestigious research factory for the Tea Research Institute (TRI) of Sri Lanka. Operating as a model facility, it combines commercial highgrown black tea manufacturing with rigorous scientific experimentation. Understanding the holistic manufacturing flow is critical before isolating the specific thermodynamic and mechanical constraints of the withering section.

1.1 From Tea Plucking to Packaging: The Production Chain

The conversion of raw green leaf into premium black tea involves a sequence of highly controlled biochemical and physical transformations. The full process chain at St. Coombs Estate is summarized as follows:

1. **Plucking:** Hand-picking of "two leaves and a bud" from the estate to ensure optimal chemical composition (polyphenols and enzymes).
2. **Withering:** Reducing the moisture content of the plucked leaf from approximately 75–80% down to 50–55% to induce physical flaccidity and trigger essential biochemical reactions.
3. **Rolling (Maceration):** Mechanically disrupting the leaf cell structures using rollers or Rotorvane machines to mix cell sap with atmospheric oxygen, initiating fermentation.
4. **Fermentation (Oxidation):** Spreading the rolled leaf on specialized tables under controlled temperature and humidity to develop characteristic colors, briskness, and aromas.
5. **Firing (Drying):** Passing the fermented leaf through a hot air dryer to arrest enzyme activity, reduce moisture to under 3%, and lock in the flavor profile.
6. **Grading & Packaging:** Sifting the dried tea into distinct particle sizes (BOP, BOPF, Dust, etc.) via mechanical sifters, followed by bulk or retail packaging in moisture-resistant materials.

2. Project-Related Focus: The Tea Withering Process

The specific core of our research project centers on the **Withering Stage**. This phase dictates the success of all subsequent manufacturing steps; improper withering cannot be corrected later in the production line. During our field visit, we mapped the sequence of steps executed by factory operators within the withering loft.

2.1 Sequential Steps in the Existing Withering System

The current operational sequence observed at the St. Coombs factory follows these specific mechanical and manual steps:

1. **Leaf Transport and Reception**

   Freshly harvested green tea leaves are brought from the estate fields directly to the upper floors of the factory (withering lofts) , and weighed to establish baseline input metrics.

2. **Initial Moisture Assessment**

   Operators perform a visual and tactile evaluation to check whether the moisture on the surface of the leaves is high or low. This assessment accounts for external weather conditions (e.g., morning dew, monsoon rain, or dry dry-weather plucking).

3. **Shedding and Trough Spreading**

   The green leaves are manually spread evenly across long, open-top withering troughs. These troughs are equipped with a perforated mesh floor through which conditioned air is blown upward from a central ducting system.

4. **Initial Thermal Inlet Adjustment**

   Based on the initial qualitative moisture level assessment, operators manually adjust the configurations of the hot air inlets (routed from the factory's boiler/drier exhaust) and the ambient air inlets. This mixing is intended to achieve the required temperature and volume for that specific trough.

5. **Active Drying Phase**

   The fans blow the mixed air stream through the bottom of the leaf bed for several hours, driving off surface and hygroscopic moisture into the atmosphere.

6. **Periodic Moisture Monitoring**

   After a designated period, operators return to check the moisture level of the leaves manually, squeezing a handful of leaves to evaluate their elasticity and flaccidity.

7. **Secondary Inlet Tuning**

   Depending on the progress of the drying and changing external weather conditions, operators manually readjust the hot air and ambient air dampers to keep temperatures within an acceptable, empirical range.

8. **Manual Leaf Turning**

   To mitigate uneven drying between the bottom layers (closer to the air jet) and top layers of the bed, laborers manually turn over the leaves using wooden rakes or hands, redistributing the mass across the trough.

Figure 1: Current manual operations observed in the St. Coombs withering loft.

3. Critical Analysis: Problems of the Existing System

While the traditional method relies heavily on the valuable experience of veteran factory operators, it introduces several significant vulnerabilities that compromise tea quality consistency, energy efficiency, and labor optimization.

1. High Subjectivity and Human Error

Moisture assessment is purely qualitative. Operators check whether moisture is "high or low" by touch and sight. This lack of quantitative data leads to frequent over-withering (shattering of leaves, loss of aromatic oils) or under-withering (excessive moisture causing high chemical loss and tearing during rolling).

2. Inability to Respond to Dynamic Weather Patterns

The ambient humidity and temperature in Thalawakelle shift rapidly throughout the day and night. Manual adjustment of air inlets cannot match these variations dynamically, leading to periods where air of improper relative humidity or excessive temperature is forced through the troughs.

3. Poor Thermal Control and High Energy Waste

Without real-time feedback loops, hot air from the drier/boiler is often over-utilized, spiking fuel wood or oil consumption. Conversely, improper mixing can send overly cold, damp air through the trough, stalling the physical wither.

4. Non-Uniform Airflow and Moisture Gradients

Static duct aerodynamics combined with variations in leaf packing density lead to localized zones of high and low pressure. Consequently, leaves located near the air entry point dry rapidly, while leaves at the far end remain wet, creating an unevenly processed batch.

Critical Technical Constraint: The Mechanical Turning Dilemma

An obvious mechanical response to the problem of uneven drying would be to establish a automated, mechanical rake or turning system along the troughs. However, **this cannot be implemented for two critical reasons:**

- **Mechanical Damage:** Rigid mechanical turners exert high shear forces that tear, bruise, and break the delicate green leaves prematurely.
- **Pre-Fermentation:** Bruising ruptures cell walls, exposing internal polyphenols to oxygen prematurely. This induces uncontrolled pre-fermentation within the withering trough, severely degrading the briskness, color, and market value of the final black tea grade.

Figure 2: Physical limitations regarding automated turning and existing duct paths.

4. Conclusion & Proposed Research Framework

Because mechanical turning of the leaves is strictly prohibited due to the risk of cell damage and prefermentation, the only viable engineering solution to achieve uniform, high-quality withering is to **control the air itself**. If the air distribution is perfectly uniform and dynamically balanced in temperature and humidity, the need for physical leaf turning is minimized or eliminated entirely.

Proposed Engineering Solution & Next Research Steps

To address these system issues scientifically, our project will proceed with the following research and design framework:

1. **Comprehensive Aerodynamic Exploration:** We must conduct an empirical research study to examine the current airflow characteristics through the main and branch ducts. This will require deploying sensors to measure key environmental and aerodynamic parameters, specifically:

    - Air Velocity and Volumetric Flow Rate (using anemometers)
    - Static and Dynamic Air Pressure (using pitot tubes/manometers)
    - Relative Humidity (RH) and Temperature Gradients across multiple points in the trough bed

2. **Automated Control System Design:** Utilizing the collected data, we will design an automated system featuring motorized dampers and a microcontroller-driven mixing chamber. This system will dynamically adjust hot air and ambient air inlets based on real-time sensory feedback.

3. **Uniform Distribution Optimization:** The duct geometry and perforated plate configurations will be modeled to guarantee uniform air velocity distribution across the entire surface area of the trough, eliminating moisture gradients without risking mechanical damage to the tea crop.